5b

y-2z).

Find

the

dimension

2.5

CO2

5a

Show

that

T:
R'’

R'
defined

by

T(,y)

=

(T+y,)

is
linear

transformation.

2.5

CO2

4b

R'(R). basis Define of

Basis

of

a
vector

space.

Show

that

the

2.5

CO2

4a

are

also

linearly

independent.

If
vj,
v2,
U3
are

linearly

independent

vectors

of
V(F),

then

show

that

v
+

v2,
V2

+

U3,
U3
+U1

2.5

CO2

3b

subspace R(R). of Let

V

= R'(R)

be

a vector

space,

then

show

that

W

=

{(a,
b,
c)
E
R:at+

b+c=

0}
is
a

2.5

CO2

integral

domain?.

Is
it

a field?.

Show

that

the

Gaussian

integer

J(i)

=

{a+

ib:

a, bE
Z}
is
commutative

ring.

ls
it
an

2.5

3a

CO1

Define

homomorphism

on

a group.

Prove

or

disprove

that

U(12)

is

cyclic

group.

2.5

2b

CO1

Define

cyclic

group.

How

many

generators

are

there

of
the

cyclic

group

G
of
order

81.

2.5

2a

CO1

abelian. Show

that

if
a,

b are

any
two

elements

of

a group

G,
then

(ab)²=

ab

if
and

only

if
G
is

2.5

lb

COI

2.5

CO1

la

li
every

element

of

a group

is
its
own

inverse,

then

show

that

the

group

must

be
abellan.

Marks

CO

No. Q

Question

and

mnentioned

in
the

answer.

Note:

All
questions

are

compulsory.

Missing

data/information

(if
any),

may

be

suitably

assumed

Maximum Marks. 25

Time:

1 hour

30
minutes

Course

title:

Mat
hematics

for
Signal

Processing

and

Communication

ECMTC04 Course code:

Mid
Semester

Examination,

September,

2022

Third

Semester-B.Tech

Course

work

no... Roll

Total

number

of
pages:

1

of
null

space

of
T.

Let

T:

R³
’

R3
be
the

linear

transformation

defined

by
T(t,

y,

z)

=

(+

2y

- z,y+2,

T t

vectors

{(1,

2, 1),
(2,1,0),

(1,
-1,2)}

form

a

Page

2 of
2

Page

1 of
2

Explain

with

the
help
of

exarmples.

What

do
you
understand

by
BIBO

Stability

of
Systems?

CO1

stable constant.

For

what

values

of
'a'
is
the

system

causal,

y()

=

J.

x(t

- a)dr

,

is
an

LTI

system?

Here

'a'
is
a
04

CO1

Determine

whether

the
system

described

by

()
An
odd
function

t
An
even

funciion

Delemine

the
vul

of

T lor
which

the
sipnal

is

01

CO1

01

=2sin x{t)

Signal.

Determine

the
Energy/

Power

of

4x[n-6).

Determine

if
x{n]

=

n uln+2

- nu•n-1]

is
an

energy

or

Power

CO1

(

Sonvider

the
lollow

ing
continuous-tne

ignal:

CO1

In
each

case

sketch

the
function.

y[n]-x[n].u[n-2]. Invariant? Time

Is
the

system

Linear,

Stable,

Causal

and

The

input

x\n]

and

u(-i+2)

04

CO1

b

CO1

1.

u(2t 6) +

period. x{n]=cos Is

cos)

periodic?

If
yes

what

is
it's

time

versions

of
the
unit

step
function

(i.e
u(tt

- to),

Express

the

following

functions

in
terms

of

shifted/flipped

No.

Questions

Marks

CO

&

Co2

Time: 1:30Hrs

Marks: Max 25

| y[0] input

is
xzn]

)
u[n
-

1].

h{n]

=

u[n

-

1).
Compute

the

system

output

when

the

the

input

x,[n]

=

)

ul-n

-

1]
and

impulse

response

01

CO2

Let

yln)

denote

the

convolution

of
h{n]

and

gín),

where

Course

Code-

ECECCOS

/ EIECCOs

Course

Title-

Signals

&
Systems

MID

Here

T
is
a

constant.

impulse

response

h(t
-

T),
find

he

output

in
terms

of
y(t).

a

3

CO2

Date

Roll No.

h(t)
produces

output

y(t)
whein

input

x(t)
is

applied.

A
Linear

Time

Invariant

systern

with

an

impuise

response

output

y[n)
of

a

system

are

related

as

mentioned thein

answer.

Attempt

al
questiors.

Missing

data

/
information

(f
any)

may
be

suitably

assumed

Compute

and

plot
the

convolution

of
x[n]

and

h[n].

Where,

1
and

y[1]

=

2,
then

find

g|l).

b h{n)
=

(1/2)n

u<n]
and

gn]=0

for
n<0

(causal

sequence).

If

EXAMINATION SEMESTER 2022

III
SEMESTER

B.Tech

(ECE,

EIOT)

When

the

integral

of
x(t)
is

applied

to
a

system

with

b)
der

the
on

en

is
wasted

through

ergency

erflow?

hyaroelectric

generator

What

is
the

robability

that

the

water

PAced

32.On

below

he

overflow

that

feeds

water

to
the

Pvents

the

depth

fron

exceeding

40.6

m.

There

is
a

pipe

There

is
an

cmergency

overflow

at
the

top

of
the

dam

that

CO2

(97

fz)=(13 5)u(x)e (/hss)

14m

is
defined

by
randon

variable

having

density

function

a

sPPosc

the

deph

of
water,

Ineasured

in
neters,

behind

the

yeu? Waf

is
the

probability

that

the

system

lifetine

will

exceed

one

Ia)-/200)a00

X<0 0sx

CO1

that

it was

manufactured

by
plant

2?

ii.
If

a relay

selected

at
random

is
found

to
be
defective,

what

is
the

probability

varit

X
defed

ss

b
Tr
fetne

of

a

system

cxpressed

in
weeks

is

a Rayleigh

random

probability

that

it
is
defective?

i.
If

a relay

is
selected

at
random

from

the

output

of
the

company,

what

is
the

2.0

CO1

erfl35j.94376

o,

= 1
Find

P[|X|

>

2].
Given

erf(1.41)=0.9533,

and

3

CO2

0 aA

rendom

variable

X
is
Gaussian

di_tributed

with

mean

=

Caice

(CDF

()
EX]

b) respectively. probabilities 0.01 and

that

a relay

manufactured

by
these

plants

is
defective

are

0.02,

0.05

producing

50,
30
and

20
percent,

respectively,

of
its

product.

Suppose

that

the

A
company

producing

electric

relays

has

three

manufacturing

plants

otherwise
1sxs2

.(i)
Find

the
constant

C

(ii)
Find

the

marginal

PDFs

fx(x)

and

fyy)

C02

otherwise x,
y >
0
and

x +y<

1

3.0

2

CO1

f,(x) 2x =

0<x< 1

Q5

below

a)
Let
X
and
Y
be
Joint

continuous

random

variables

with

joint

PDF

is
given

(ii)
that

no

calls

will
arrive

during

any
10
minute

period

b
Le
X
s

a random

variabie

and
its
PDF

is
given

by

Find

moment

generating

function

of
X.

(i)
that

more

than

three

calls

will
arrive

during

any
10
minute

period

3

CO1

otherwise 0

3

CO3

probability period

is
known

to
be
Poisson

random

variable

X
with

parameter

A
=2.
Find

the

b)
The

number

of
telephone

calls

arriving

at

a switchboard

during

any
10
min

P(X-0|X< Find 0.5)

A
continuous

random

variable

X
has

the

PDF

Questions

Marks

CO

Time

1
30
Ers

Terthored arswe the Course

Cod

ECECCO6

EIECC06

Attempt

alluestos

Missing

datainfoation

(1f
any)

may

be

suitably

assumed

&

MID-

SENMESTER

EXAMINATION,

September

2022

THIRD

SEMESTER-BTECH.(ECE

& EIOD)

Course

Title-

Probability

Theory

and
Stochastic

Process

Marks: Max 25

0,
x

CO2

Q

a)
Let

X
be

a random

variable

with

cdf

(i)

P (X<
1/4)

(ii)

P (0
< X<

1/4)

Find

.......X 21/2 I...

F(X) =

Date

Roll No.

5.

.... 0Sx

<1/2

..X <0
<

1,x >1
0Sx <1

4

(a)

2(b)

1b)

2a)

Course Code: ECECC07
Course Title: Microelectronics
Time 01 80 Hre.

Qla)

Max Marka: 16 Note: Attempt all the five questions, data in formation if any ehould be suitably assumed and mentioned in the answer.

A) queationa carry equal marks. Missing

No.

MID SEM EXAMINATION, sept 2022

ech. (ECE)

For the differentil amplifier shown in ig 1, find A, and Acm

loo kR

DD

2K

2cok)E2mA

ii) Find overall gain

in

Define
Fig.

CMRR
1.

for differential amplifier and calcuiate the same for the circuit shown

For the circuit shown in Fig. 3

A,V,).

In CE amplifier shown in Fig.2, the parameters are B =100, Ico2.3 mA, ro=0, Find
Re

i) Find Rin for the circuit shown in Fig. 2.

Question

2k

Vce

i) Draw small signát cquivalent circuit.

Fig:2

ii) Explain how voltage gain wil! be affected when a resistance is added in emitter of Fig. 2.

B00

Fig. 1.

Fig. 3

Find input impedance (Rin) of the circuit given in Fig. 3.

Draw the circuit diagram of Darlington pair using BJT and also draw its smail signal equivalent circuit.

Mar
ke

(2)

(1)

(2)

(1)

(1)

CO

(2)

CO2

CO1

(2) CO2

COI

CO2

CO2

4(b)

5(a)

Find input impedance of Darlington pair. For the circuit shown in Fig. 4, B-100, VA 100 V, lco-I mA, C,= 19.5 pF and

C-0.5 plF.

Draw small signal high frequency equivalent circuit.

5(6)Find fy of the circuit shown in Fig. 4.

Rezsk

Fi4

(1) C02

CO3

) CO3

Netaji Subhas University of Technology Total number of Pages.

MID Scmester of IIIrd Semester B.Tech- ECEloT

Course Code-MID-SEMESTER EIECCO7 EXAMINATION, SEPTEMBER, 2022
Time-

Course
1.5

Title-

Hour Microelectronics Circuits and Applications

Q. No
la

Note:- Attempt all questions. Missing assumed &mentioned in data/information (if any), may be suitably the answer.

Question To design active-loaded differential amplifier as shown in fig. I to obtain differential gain of 50OV/V. The technology available provides unCox= 4upCox=400uA/N,|VA|=10, L=0.5um, | Vt| =0.5V,and operates from t1V supplies. Use a bias current I=200LA and operate all devices at |vGS-Vt|=0.2V. (a) Find the W/L ratios of the four transistors.

Fig.1

Vps

Roll no.

For the Fig. 1 of guestion (1), If t is delivered by a simple NMOS current source operated at the samne VGS-Vt and having the same channel length as the other four transistors, determine the CMRR obtained.

1.5

Marks
1.5

Max.Marks- 15

CO2

COI

CO

2a

26
3a
3b

4a
4b

5a

Sb

Rg

R

R

Q

Fig.2
For lo = 2Am, Ro 50 kQ, Rg = 1k2, R =100 , Rc = 10 kn, V= 20 V, V-=-20 V, V,= 0.025 V, r, = 20 2, B = 99, Vae =0.65 V, and VA = 50 V, calculate Vo, Vo2. Find rout and CMRR for the Fig.2 shown in Question.2a Explain any three biasing techniques in Amplifiers. For the high frequency equivalent circuit of a common source MOSFET amplifier shown in Fig.3 having Rsig- 100KO, Ru-4.7M2, Cgs=l pF, Cgd -0.4pF, gm= ImA/V and RL= Rp= 15KQ, Find the midband

Vo gain AM: and upper 3-dB frequency

Fig.3
What is the Miller's theorem, explain in detail. Consider CG amplifier circuit which has gm= ImA/V, Rsig 502, Ry 1SKS, Ry= 15KQ. Find Rin, Rout, Av,Avo, and Gv
For CS stage with resistive load amplifier prove that Gain =-gmrd
Explain the operation of a MOS differèntial pair with common mode input voltage.

I.5

1.5
1.5
1.5

1.5
1.5

1.5

1.5

CO2

CO2
CO1
CO1

COI
COI

COI

C02

b)
er

he

n

n

2

is
wasted

through

ergency

erilow?

hydroelectric

generator.

What

is
the

probability

that

the

water

placed

32.0m

below

the

overflow

that

feeds

water

to
the

prevents

the

depth

from

exceeding

40.6

m.

There

is
a

pipe

There

is
an

emergency

overilow

at-
the

top

of
the

dam

that

3

CO2

-(/135)u(x)e-has) fa

Q3

-d£m

is
defined

by
random

variable

having

density

function

a)
Süppose

the

depth

of
water,

measured

in
meters,

behind

the

year? What

is
the

probability

that

the

system

lifetime

will

exceed

one

x<0

A)=}20)le-*a

2

CO1

0<x

variable

X
defined

as

b) The
lifetime

ofa

systemn

expressed

in
weeks

is

a Rayleigh

random

erf(1.35)=0,94376 0

ahd

Oy
=1.

Find

P[|X\

> 2].
Given

erf(1.41)=0.9533,

and

3

CO2

Q2

a)
A
random

variable

X
is
Gaussian

di_tributed

with

mean

=

Calculate:

()
CDF

(ii)
E[X]

otherwise
1<x< 2

=2x? f(x)

2

CO1

0<x<1

b)
L
X
is

a random

variable

and
its
PDF

is

given

by:

Find

moment

generating

function

of
X.

otherwise 0,

3

CO3

(6)

e/3 >0 x
A
continuous

random

variable

X
has

the

PDF

Marks

CO

No. Q.

Questions

mentioned

i
the
answer.

Attempt

all
questions.

Missing

data

/ information

(|f
any)

may

be

suitably

assumed

&

Marks: Max25

ECECCO6/EIECC06 Time: 1:30Hrs Course Code-

MID-SEMESTER

EXAMINATION,

September

2022

THIRD

SEMESTER-B.TECH.(ECE

&

EIOT)

Course

Title-

Probability

Theory

and
Stochastic

Process

that

it
was

manufactured

by
plant

2?

ii.
Jf

a relay

selected

at

random

is
found

to
be

defective,

what

is
the

probability

probability

that

it

is defective?

i.
If
a

relay

is
selected

at

random

from

the

utput

of
the

company,

what

is
the

respectively. probabilities 0.01 and

thata

relay

manufactured

by these

plants

is defective

are

0.02,

0.05

producing

50,
30
and
20

percent,

respectively,

of
its

product.

Suppose

that

the)

¤

company

producing

electric

relays

has

three

manufacturing

plants

2.0

b)

COI

()
Find

the
constant

C

(ii)
Find

the

marginal

PDFs

fx(x)

and

fy(y)

frv(x,
y)

= {C+1

otherwise X,
y >0

and

x +y<1

C02

Q5

below

a)
Let
X
and
Y
be
Joint

continúous

random

variables

with

joint
PDF

is

given

3.0

(ii)
that

no

calls

will
arrive

during

any
10
minute

period

(i)
that

more

than

three

calls

will
arrive

during

any
10
minute

period

probability period

is
known

to
be
Poisson

random

variable

X
with

parameter

=2.
Find
the

b)
The
number

of

telephone

calls

arriving

at
a

switchboard

during

any
10
min

3

CO1

P(X-0X< Find 0.5)

(x) Fx =

0,x <0)27.0Sx<1 1

CO2

Q4

a)
Let
X
be

a random

variable

with

cdf
1,
x
>

1

()

P (Xs
1/4)
(i)
P

(0
< X<
1/4)

Find

x21/2

F(x) =

.0
Sx
<

1/2

1 0

*..X0

Page of2 1

b)
State

the

axioms

of

Piobability.

01

CO1

v.

P(AB)

=

P(A}P(B)

P(B)

p(A)

iv,

P(AJB)

P(B4)

ü.

PA)=0,

or

PB)

=

0,
or

both.

2

ü.

P(AUB|C)

= P(4)

+ P("/c)

04

CO1

i.

P()= P(4)

reasons.

a)
Events

A
and

B
are

mutually

exclusive.

Determine

which

of
the

following

are
true

and

which

are

false.

Explain

the

Plot

the
CDF

F()

for
every

x

from

-o
to
o.

X(H)=

1 and

X(T).

: 0.

01

CO2

b) 'q'.
We
define

a

random

variable

X
such

that

heads

(H)
equals

p'
and

the

probability

of
tails

(T)
equals

In
the

coin

tossing

experiment,

the

probability

of

UB|C) P(A iv. ii.

P(A

U B|B)

iü.

P(AU B)

P(AB) i.
P(B) =

and b

probabilities

in
terms

(f
a,
b
and

c.

P(C) C. =

a)
Events

A,
B
and

C
are

04

CO11

No.

Questions

Marks

CO

ssumed

&
mentioned

in
the
answer.

Atterpt

all
questions.

Missing

data

/
information

(f
any)

may

be
suitably

Time: 1:30 Hrs

Marks: Max 25

-INECCO6/1TECCO6 CourseCode

Course

Title-Probability

%

Stochastic ProcesSes

MID-SEMESTER EXAMINATION 2022
I

Date:

Roll No.

independent,

with

P(A)

=

a,

Determine

following the

SEMESTER

R.Tech ITNS) (IT,

*****

Y=

aX+b

have

mean

variance ZERO and

CO2

Compute

the

mean

score.

elsewhere

5

fg(r)

=(1

+ r)

for

r
>

0

the
hit
from

the

centre.

SuppOse

that

the
PDF

of
R
is

Let

R
be
the

random

variable

.representing

distance

of

third

ring

2

points.

Shots

outside

of
the

target

count

.

points,

within

the

next

ring

3

points,

and

within

the

a

A
target

is
made

of
three

concentric

circles

of
radii

3
7,1,
and

3/2

feet.

Shots

within

the

inner

circie

count

4

2

04

CO2

b) Express

P[|Y<y]

in
terms

of
the
CDF

of
Y.

01

CO2

this

is
called

the

memoryless

property.

iiü.
Show

that

P{X

>t+x|X

>t}=

P[X

>

x].
Explain

why

ü. Find

and

plot

fr(x|X

>

t).

from (r)? F, Find

and

plot

Fx(r|X

>

t).
How

does

F,(x|X

>

t)
differ

04

CO2

a)
Let
X
be
the

exponential

random

variabie.

P-/, in

in
terms

of k

ard

01

3

CO2

for

x
>

0

for

x

< 0

04

1

CO2

a)

A rAndom

variable

X
has

CDF

given

as

ONE?

b)
Let

the

mean

ari

variance

of
X
be

m

and

g
respectively.

For

what

values

of

a

and

b
does

the

random

variable

01

A.

ii.
FindP[N<

2).
P(:0).

P[X<0].

P{2
<X<6).

P[X

> 101

with

parameter

À if
X
takes

on

values

0,
1,
2,
...n,

Find

Plot the i.

CDF

And

identify

the

type

of
random

variable

b)
State

the

Poisson

distribution

for

random

variable

X

3

4

5

2

Total no. of pages 1

B.Tech. II Semester (ECE/EloT)

1

Course Code: ECECC08/EIECCO08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

MID-SEMESTER EXAMINATION, SEPTEMBER 2022

Time: 1:30 Hours

Q.No. Question

mentioned

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and in the answer.

Simplify the following Boolean function F with the don't-care conditions d and express in the SOP form

(a) Express in the SOP form

F(A, B, C, D) = £ (5, 6, 7, 12, 14, 15) and d(A, B, C, D) = (3,9,11)

(b) Express in the POS form

(b) If X=1 in the logic equation

(a) Implement the expression obtained in QI using NOR gates only.

of Z.

[X+Z{Y+(Z'+XY"}I{X+Z(X+Y)}=1, then determine the value

Roll no.

Reduce the following Boolean expressions using boolean laws
(a) F (w.x,yz)= (xy + z) tz+xy + wz
(b) F(4,B. C,D) = AB(D + CD) + B(A + ACD)

Perform subtraction on the given numbers

Max. Marks: 15

(a) 100010.001 - 111.11 using the 2's complement (b) 163.25 - 745 using the 9's complement
(a) Drawa basic CMOS inyerter circuit.
(b) Find the Maxterms of the following expression F(w,x,y,z) = (w+x')Xw'+ytz')\xty'+z')

Marks

1.5+ CO1
1.5

1.5+
1.5

1.5+
1.5

1.5+
1.5

1.5+

CO

1.5

CO3

CO1

CO1

CO1

CO2

CO1

CI)FN) 23|4)22

(35)

Roll n.

Total

no.

of

pages

1

B.Tech.

II

Semester

(EE/CE)

MID-SEMESTER

EXAMINATION,

SEPTEMBER

2022

Course

Code:

ICECCO8/

EEECC08

Course

Title:

DIGITAL

CIRCUITS

AND

SYSTEMS

Max,arks: 25

Time: 1:30 Hours

Note:

Attempt

all

questions.

Missing

data/information

(if

any),

may

be

suitably

assumed

and

mentioned

in

the

answer.

CO

Marks

Question

Q.No.

CO1

2.5+ 2.5

Simplify

the

following

Boolean

function

F

with

the

don't-care

conditions d

and

express

in

the

SOP

form

F(A,

B,

C,

D) =

2

(5,

6,

7,

12,

14,

15)

and

d(A,

B,

C,

D) =

2

(3,9,11)

(a)

Express

in

the

SOP

form

(b)

Express

in

the

POS

form

CO3

2.5+

(a)

Implement

the

expression

obtained in

Q1

using

NOR

gates

only.

2

2.5

CO1

(b)

IfXl in

the

logic

equation
[X+Z{Y'+(2'+XY'")}J{

X+Z'(X+Y)}=1,

then

determine

the

value

of Z.

CO1

2.5+ 2.5

Reduce

the

following

Boolean

expressions

using

boolean

laws

(a)

F(wx,yz)=

(xy

+

z)

tz+xy +

wz

3

(b)

F(4,B,

C,D)

=

AB(D +

CD) +

B(A

+

ACD)

CO1

2.5+

Perform

subtraction

on

the

given

numbers

4

2.5

(a)

100010.001 -

1l1.11

using

the

2's

complement

(b)

163.25 -

745

using

the

9's

complement

CO2

2.5+

(a)

Draw a

basic

CMOS

inverter

circuit.

5

2.5

CO1

(b)

Find

the

Maxterms

of

the

following

expression

(wtx')(w'tytz')X(*+y'+z') = F(w,x,y,z)